

SYDNEY TECHNICAL HIGH SCHOOL

YEAR 11 YEARLY EXAMINATION

MATHEMATICS EXTENSION 1

2005

Time allowed: 90 minutes

Directions to Candidates

- Attempt all questions
- Start each question on a new page
- All necessary working should be shown
- Unless otherwise specified, answers must be given in their simplest form
- Approved calculators may be used in all sections.
- Use a ruler when drawing straight lines
- Marks may be deducted for careless or poorly arranged work.
- Marks shown are approximate and may be varied.

Name: _____ Class: _____

1	2	3	4	5	6	TOTAL

Question 1

	Marks
a) Express in simplest form $6 \times 3^n + 3^{n+1}$.	2
b) If $(x - 2)$ is a factor of $P(x) = x^3 + ax + 2$, find the value of a .	1
c) Form a quadratic equation with roots $1 - \sqrt{5}$ and $1 + \sqrt{5}$. Give your answer in general form.	2
d) Find the acute angle between the lines $x = 3$ and $y = \frac{1}{2}x + 1$ to the nearest minute.	2
e) If α, β, γ are the roots of the equation $x^3 - 2x + 5 = 0$. Find the value of $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$.	2
f) Draw a sketch of $P(x) = (x + 3)(x - 2)^2$ clearly indicating the x and y intercepts	2

Question 2

a) Let A and B be the points $(0,1)$ and $(2,3)$ respectively.	5
i) Find the equation of the perpendicular bisector of AB .	
ii) The point P lies on the line $y = 2x - 9$ and is equidistant from A and B . Find the coordinates of P .	
b) Find the quotient when $P(x) = 2x^3 + 3x^2 - 8x - 17$ is divided by $T(x) = x^2 - 4$.	3
c) Given $\sin 2\theta - \tan \theta \cos 2\theta = \tan \theta$. Find the exact value, in rationalised form, of $\tan 67\frac{1}{2}^\circ$.	3

Question 3

a) i) Sketch the graph of $x^2 + y^2 = 25$	5
ii) Explain why $x^2 + y^2 = 25$ is not a function.	
iii) By choosing the appropriate function, find the gradient of the tangent at the point $(3,4)$ on the curve $x^2 + y^2 = 25$.	

- b) Consider the parabola $8x = y^2 + 4y + 12$ 4
- i) By completing the square find the coordinates of the vertex.
- ii) Sketch the parabola showing the vertex, focus and any intercepts.
- c) $P(x)$ is an odd monic polynomial of degree 3 with 2 as a zero. Write down the equation of $P(x)$. 2

Question 4

- (a) i) If $x^4 - 6x^3 + 7x^2 + 6x - 8 \equiv (x^2 + ax)^2 + b(x^2 + ax) + c$, find the numerical values of a , b and c . 4
- ii) Hence or otherwise, solve the equation $x^4 - 6x^3 + 7x^2 + 6x - 8 = 0$.
- b) The points $P(12t, 6t^2)$ and $Q(36, 54)$ are points on a parabola 3
- i) Find the Cartesian equation of the parabola
- ii) If PQ is a focal chord find the value of t .
- c) i) For what value of k does $x^2 - kx - k = 0$ have real roots 4
- ii) Hence or otherwise find the range of $y = \frac{x^2}{1+x}$.

Question 5

- a) i) Derive the equation of the tangent to the parabola $x = 2t, y = t^2$ at the point P , where $t = p$. 7
- ii) If Q is the point on the parabola where $t = q$, and OQ is parallel to the tangent at P (O is the origin), show that $q = 2p$.
- iii) M is the midpoint of PQ . Find the co ordinates of M .
- iv) If P and Q move along the parabola so that OQ always remains parallel to the tangent at P , show that the equation of the locus of M is $5x^2 = 18y$.
- b) The formula for the area of a rectangle is given by $A = 8x \sin \theta - x^2 \tan \theta$ where θ remains fixed 4
- i) By treating A as a quadratic function show that the maximum value of A is $A = 8 \sin 2\theta$

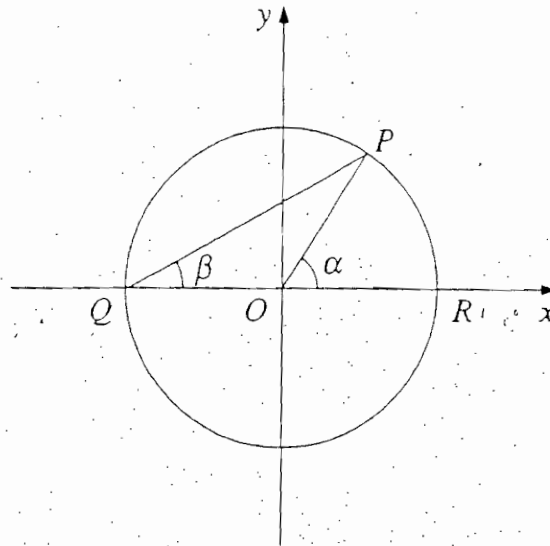
Question 6

a) The equation $|x^2 - 4x| = k$ has 3 solutions. Find the value of k .

1

b)

10



In the diagram, Q is the point $(-1, 0)$, R is the point $(1, 0)$, and P is another point on the circle with centre O and radius 1. Let $\angle POR = \alpha$ and $\angle PQR = \beta$, and let $\tan \beta = m$.

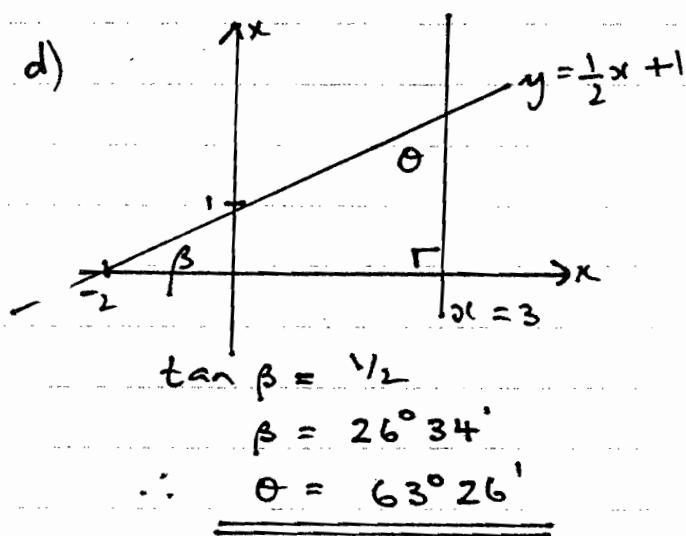
- (i) Explain why $\triangle OPQ$ is isosceles, and hence deduce that $\alpha = 2\beta$.
- (ii) Find the equation of the line PQ in terms of m .
- (iii) Show that the x coordinates of P and Q are solutions of the equation $(1 + m^2)x^2 + 2m^2x + m^2 - 1 = 0$.
- (iv) Write an expression for the sum of the roots of this quadratic equation
- (v) Hence or otherwise find the coordinates of P in terms of m .
- (vi) By using a right angled triangle show that $\tan 2\beta = \frac{2 \tan \beta}{1 - \tan^2 \beta}$

QUESTION 1

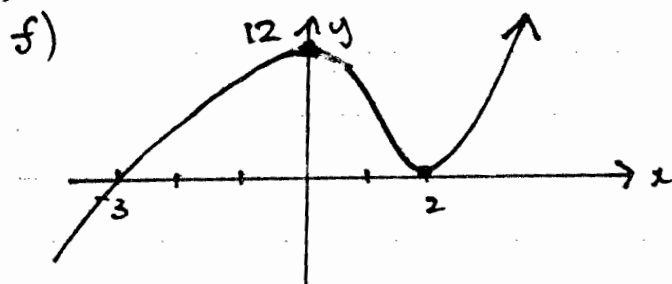
$$\begin{aligned}
 \text{a)} \quad 6 \cdot 3^n + 3^{n+1} &= 6 \cdot 3^n + 3 \cdot 3^n \\
 &= 3^n(6+3) \\
 &= 9 \cdot 3^n \\
 &= 3^2 \cdot 3^n \\
 &= \underline{\underline{3^{n+2}}}
 \end{aligned}$$

$$\begin{aligned}
 \text{b)} \quad P(2) &= 0 \\
 8 + 2a + 2 &= 0 \\
 2a &= -10 \\
 \underline{\underline{a = -5}}
 \end{aligned}$$

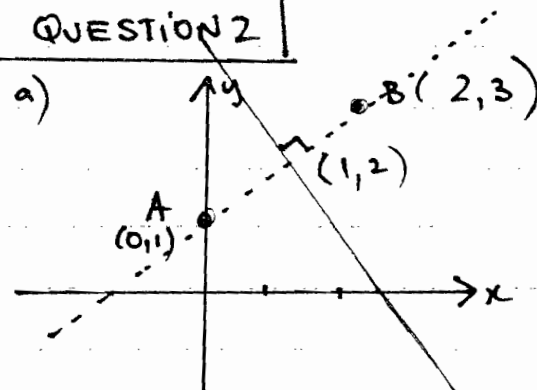
$$\begin{aligned}
 \text{c)} \quad x^2 - (\text{sum roots})x + \text{product} &= 0 \\
 \underline{\underline{x^2 - 2x - 4 = 0}}
 \end{aligned}$$



$$\begin{aligned}
 \text{e)} \quad \frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} &= \frac{\beta\gamma + \alpha\gamma + \alpha\beta}{\alpha\beta\gamma} \\
 &= \frac{c/a}{-d/a} \\
 &\rightarrow = \underline{\underline{\frac{-2}{-5}}}
 \end{aligned}$$



QUESTION 2



i) midpt AB (1, 2)

$$m_{AB} = \frac{2}{2} = 1 \quad \therefore \text{perp } m = -1$$

eqn perp bisector $y - 2 = -1(x - 1)$

$$\underline{\underline{x + y - 3 = 0}}$$

ii) $P(x, 2x - 9)$ lies on $x + y - 3 = 0$

$$x + 2x - 9 - 3 = 0$$

$$3x - 12 = 0$$

$$3x = 12 \quad \therefore x = 4$$

$$\underline{\underline{\therefore P(4, -1)}}$$

b)

$$\begin{array}{r}
 2x + 3 \\
 x^2 - 0x - 4 \overline{) 2x^3 + 3x^2 - 8x - 17} \\
 \underline{2x^3 + 0x^2 - 8x} \\
 3x^2 + 0x - 17 \\
 \underline{3x^2 + 0x - 12} \\
 -5
 \end{array}$$

$$c) \sin 2\theta - \tan \theta \cdot \cos 2\theta = \tan \theta$$

$$\sin 2\theta = \tan \theta + \tan \theta \cos 2\theta$$

$$\sin 2\theta = \tan \theta (1 + \cos 2\theta)$$

$$\frac{\sin 2\theta}{1 + \cos 2\theta} = \tan \theta$$

$$\therefore \tan 67\frac{1}{2}^\circ = \frac{\sin 135^\circ}{1 + \cos 135^\circ}$$

$$= \frac{\sin 45^\circ}{1 - \cos 45^\circ}$$

$$= \frac{1}{\sqrt{2}} \div \left(1 - \frac{1}{\sqrt{2}}\right)$$

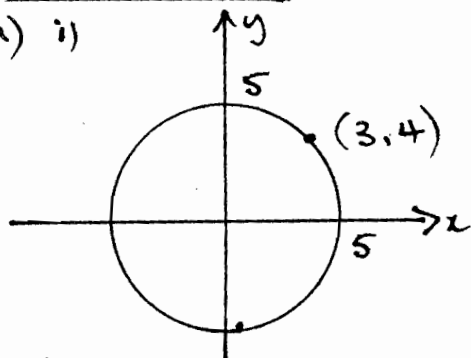
$$= \frac{1}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}-1}$$

$$= \frac{1}{\sqrt{2}-1} \times \frac{\sqrt{2}+1}{\sqrt{2}+1}$$

$$= \underline{\underline{\sqrt{2}+1}}$$

QUESTION 3

a) i)



ii) Vertical line cuts more than once between
 $x = -5$ & $x = 5$

$\therefore$ each x value will not have a unique y value.

$$\frac{dy}{dx} = \frac{-x}{\sqrt{25-x^2}}$$

at $x = 3$ $m = \underline{\underline{-\frac{3}{4}}}$

b) i) $y^2 + 4y = 8x - 12$

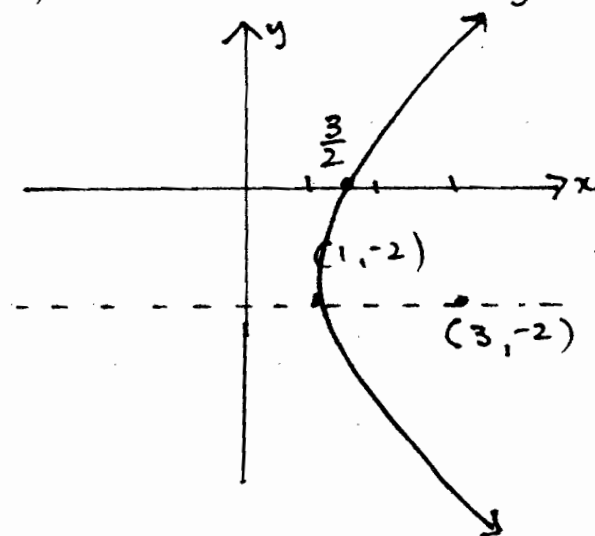
$$(y^2 + 4y + 4) = 8x - 12 + 4$$

$$(y + 2)^2 = 8x - 8$$

$$(y + 2)^2 = 8(x - 1)$$

Vertex $\underline{\underline{(1, -2)}}$

ii) $4a = 8 \therefore a = 2$ if $y = 0 \Rightarrow x =$



c) $P(x) = x^3 + ax + b$

$P(0) = 0$ passes through origin

$P(2) = 0$ given

$$\therefore b = 0 \quad 8 + 2a = 0$$

$$2a = -8$$

$$a = -4$$

$$\therefore P(x) = \underline{\underline{x^3 - 4x}}$$

$$= x(x^2 - 4)$$

QUESTION 4

a) i)

$$x^4 - 6x^3 + 7x^2 + 6x - 8$$

$$\equiv x^4 + 2ax^3 + a^2x^2 + bx^2 + abx + c$$

$$2a = -6 \quad a^2 + b = 7 \quad ab = 6 \quad c = -8$$

$$\underline{a = -3} \quad 9 + b = 7 \quad \checkmark$$

$$\underline{b = -2} \quad \underline{c = -8}$$

ii) $(x^2 - 3x)^2 - 2(x^2 - 3x) - 8 = 0$

Let $u = x^2 - 3x$

$$u^2 - 2u - 8 = 0$$

$$(u - 4)(u + 2) = 0$$

$$u = 4 \quad u = -2$$

$$x^2 - 3x = 4 \quad x^2 - 3x = -2$$

$$x^2 - 3x - 4 = 0 \quad x^2 - 3x + 2 = 0$$

$$(x - 4)(x + 1) = 0 \quad (x - 2)(x - 1) = 0$$

$$\therefore x = 4, -1, 2, 1$$

b) i) $x = 12t \quad y = 6t^2$

$$\frac{x}{12} = t$$

$$\therefore y = 6\left(\frac{x}{12}\right)^2$$

$$\underline{y = \frac{x^2}{24}}$$

ii) PQ passes throu' (0,6) focus

$$m_{PQ} = \frac{54 - 6t^2}{36 - 12t} = \frac{6(9 - t^2)}{12(3 - t)}$$

$$= \frac{6(3 - t)(3 + t)}{12(3 - t)}$$

$$= \frac{3 + t}{2}$$

= eqn PQ

$$y - 54 = \frac{(3 + t)(x - 36)}{2}$$

sub pt (0,6)

$$6 - 54 = \frac{(3 + t)(-36)}{2}$$

$$-48 = -18(3 + t)$$

$$\frac{48}{18} = 3 + t$$

$$\underline{t = -1/3}$$

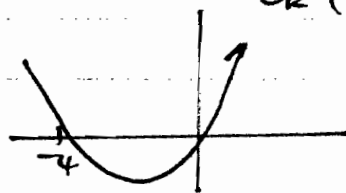
c) i) $\Delta \geq 0 \quad k^2 - 4 \cdot 1 \cdot (-k) \geq 0$

$$k^2 + 4k \geq 0$$

$$k(k + 4) \geq 0$$

$$\therefore k \geq 0$$

$$\underline{k \leq -4}$$



ii) $y = \frac{x^2}{1+x}$

$y \neq 0$
Let $y = k$

$$k(1+x) = x^2$$

$$x^2 - xk - k = 0$$

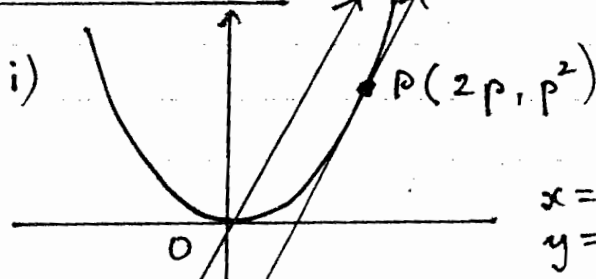
solutions for k if $\Delta \geq 0$

$$\therefore k \geq 0 \quad k \leq -4 \text{ from above}$$

$$\therefore \underline{\text{Range } y \geq 0, y \leq -4}$$

QUESTION 5

a) i)



$$x = 2t$$

$$y = t^2$$

$$y = \frac{x^2}{4}$$

$$\frac{dy}{dx} = \frac{2x}{4} = \frac{x}{2}$$

$$Q(2q, q^2)$$

$$m_{\frac{PQ}{P}} = \frac{2q}{2} = q$$

tang: $y - q^2 = q(x - 2q)$

$$y - q^2 = qx - 2q^2$$

$$y = qx - q^2$$

ii) gradient of OQ = $\frac{q^2}{2q} = \frac{q}{2}$

since OQ || to tangent at P

$$\frac{q}{2} = q \therefore q = 2p$$

iii) $M(P+q, \frac{P^2+q^2}{2})$

iv) $x = P+q \quad y = \frac{P^2+q^2}{2}$

$$(P+q)^2 = P^2 + q^2 + 2Pq$$

$$2y = P^2 + q^2$$

$$x^2 = 2y + 2(P \cdot 2P)$$

$$x^2 = 2y + 4P^2$$

since $2y = P^2 + q^2$

$$2y = P^2 + (2P)^2$$

$$2y = 5P^2$$

$$\frac{2y}{5} = P^2$$

$$\therefore x^2 = 2y + 4\left(\frac{2y}{5}\right)$$

$$5x^2 = 10y + 8y$$

$$5x^2 = 18y$$

b) i) $A = 8x \sin \theta - x^2 \tan \theta$
axis of sym: $x = \frac{-8 \sin \theta}{-2 \tan \theta}$

$$x = \frac{4 \sin \theta \cdot \cos \theta}{\sin \theta}$$

$$x = 4 \cos \theta$$

$$\therefore \max A = 8 \cdot 4 \cos \theta \sin \theta - (4 \cos \theta)^2 \tan \theta$$

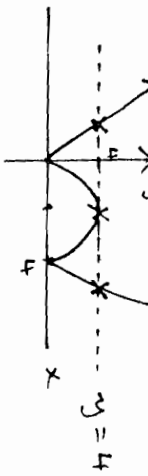
$$= 32 \cos \theta \cdot \sin \theta - 16 \cos^2 \theta \cdot \tan \theta$$

$$= 32 \cos \theta \cdot \sin \theta - 16 \cos \theta \cdot \sin \theta$$

$$= 16 \cos \theta \cdot \sin \theta$$

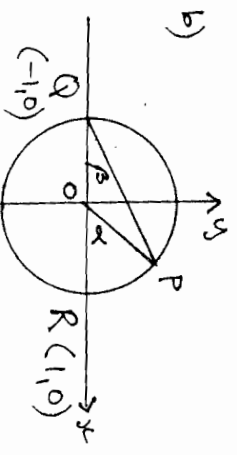
QUESTION 6

a) $y = |x^2 - 4x|$ sketch



$$\therefore |x^2 - 4x| = k \text{ has 3 solutions}$$

$$\text{if } k = 4$$



i) $OP = OQ$ (equal radii)

$\therefore \triangle OPQ$ is isosceles and $\angle POQ = \beta$ (opp. equal sides)

$$\therefore \alpha = 2\beta \text{ (ext. angle of } \triangle)$$

ii) eqn of PQ gradient = m
and $Q(-1,0)$

$$y - 0 = m(x + 1)$$

$$y = mx + m$$

iii) eqn circle centre (0,0) radius 1

$$x^2 + y^2 = 1$$

PQ: $y = mx + m$ } sim. eq

$$x^2 + (mx + m)^2 = 1$$

$$x^2 + m^2 x^2 + 2m^2 x + m^2 = 1$$

$$x^2(1 + m^2) + 2m^2 x + m^2 - 1 = 0$$

iv) Sum Roots = $\frac{-2m^2}{1+m^2}$
 $(\alpha + \beta)$

v) if one root is -1

let other root be α

$$\alpha - 1 = \frac{-2m^2}{1+m^2}$$

$$\frac{1+m^2}{1+m^2}$$

$$\alpha = \frac{-2m^2}{1+m^2} + 1$$

$$\alpha = \frac{-2m^2 + 1 + m^2}{1+m^2}$$

$$\alpha = \frac{1-m^2}{1+m^2}$$

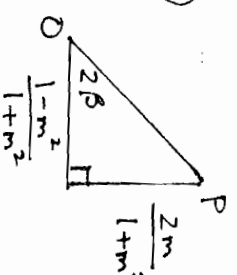
sub into

line PQ

$$y = m\left(\frac{1-m^2}{1+m^2}\right) + m$$

$$\therefore P\left(\frac{1-m^2}{1+m^2}, \frac{2m}{1+m^2}\right)$$

vi)



$$\tan 2\beta = \frac{\frac{2m}{1+m^2}}{\frac{1}{1+m^2}}$$

$$\frac{2m}{1+m^2}$$

$$\tan 2\beta = \frac{2m}{1+m^2}$$

$$m = \tan \beta$$

$$\tan 2\beta = \frac{2 \tan \beta}{1 - \tan^2 \beta}$$

